Regression Analysis

Fitting the least-squares line, regression coefficients, the two regression lines and model assumptions.

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1 The model

Regression models how a **dependent** (response) variable depends on one or more **independent** (predictor) variables.

Simple linear regression

$$y_i = \alpha + \beta x_i + e_i$$

Error e_i

Theoretical deviation from the true line

Residual $\hat{e}_i$

Observed − fitted: $y_i - (\hat{\alpha} + \hat{\beta} x_i)$

2 Slope & intercept

- **Slope** β: expected change in y for a one-unit increase in x. Range $-\infty$ to $+\infty$; $+$ direct, $-$ inverse.
- **Intercept** α: value of y when $x = 0$.

Simple	1 predictor
Multiple	2 or more: $y = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k + e$
Logistic	Categorical response

3 Least squares

Choose α, β to **minimise** $E = \sum [y_i - (\alpha + \beta x_i)]^2$. This gives the **normal equations**:

First

$$\sum y = n\alpha + \beta \sum x$$

Second

$$\sum xy = \alpha \sum x + \beta \sum x^2$$

Slope

$$\hat{\beta}_{yx} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

Intercept

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Equivalently $\hat{\beta}_{yx} = cov(x, y)/var(x)$. The fitted line passes through $(\bar{x}, \bar{y})$.

4 Two regression lines

y on x (predict y)

$$\beta_{yx} = r \frac{\sigma_y}{\sigma_x}$$

x on y (predict x)

$$\beta_{xy} = r \frac{\sigma_x}{\sigma_y}$$

Link to correlation

$$\beta_{yx}\, \beta_{xy} = r^2$$

5 Properties of regression coefficients

- ✓ Both have the **same sign**, and r takes that sign: $r = \pm\sqrt{\beta_{yx}\beta_{xy}}$.
- ✓ If one exceeds 1, the other must be less than 1.
- ✓ $r = 0$: both coefficients are zero.
- ✓ $r = \pm 1$: the two lines **coincide**.
- ✓ Independent of change of **origin**, not of **scale**: with $X = \frac{x-a}{h}$, $Y = \frac{y-b}{k}$, $\beta_{YX} = \frac{h}{k}\beta_{yx}$.

6 Assumptions

- **Linearity** of the mean relationship
- **Independence** of errors (no **autocorrelation**)
- Constant error variance: **homoscedasticity**
- **Normality** of errors, for exact inference
- No exact **multicollinearity** among predictors
- Correct model specification

Check the fitted model with residual plots.

7 Correlation vs regression

	Correlation	Regression
Purpose	Measures association	Models the relationship and predicts
Symmetry	Symmetric: $r_{xy} = r_{yx}$	Asymmetric: y on $x \neq x$ on y
Units	Unit-free	Coefficients carry units
Range	−1 to +1	Unbounded

! Common mistakes

- × Using the y-on-x line to predict x.
- × Coefficients of opposite signs (impossible).
- × Treating a regression fit as proof of cause.
- × Predicting far outside the data range.

? Fill in the blanks

1. With ŷ = 5 + 2x, the predicted y at x = 10 is ———. [25]
2. If x̄ = 4, ȳ = 13 and the slope is 2, the intercept is ———. [5]
3. If b_yx = 0.8 and b_xy = 0.45, then r = ———. [0.6]
4. If b_yx = 1.6, then b_xy cannot be more than ———. [0.625]
5. If b_yx is negative, b_xy must also be ———. [negative]
6. When r = 0, the two regression lines are ——— to each other. [perpendicular]

Source: *Textbook of Agricultural Statistics*, Ch. 9 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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